Adaptive Acoustic Vehicle Alerting System (AVAS) for Urban Indian Conditions

Aditya Sinha, Ayanjyoti Roy, Pranshu Raj

Abstract—**Electric vehicles (EVs) produce significantly less noise than internal combustion engine vehicles, particularly at low speeds. While this reduces noise pollution, it creates safety challenges for pedestrians who rely on auditory cues to detect approaching vehicles. Acoustic Vehicle Alerting Systems (AVAS) generate artificial warning sounds to improve pedestrian awareness. However, most AVAS implementations use fixed sound characteristics and do not adapt to environmental conditions. This work presents an adaptive AVAS prototype designed for Indian urban environments. The proposed system integrates ambient noise sensing, pedestrian proximity detection, and rule-based sound generation. MATLAB/Simulink simulation and a hardware prototype demonstrate the feasibility of adaptive acoustic alerts for improving pedestrian safety in dense urban traffic environments.**

I. Introduction

Electric vehicles are becoming increasingly common due to their environmental benefits and energy efficiency. However, EVs are significantly quieter than traditional vehicles at low speeds because they lack internal combustion engines.

This silent operation creates safety risks for pedestrians, cyclists, and visually impaired individuals who rely on auditory cues to detect nearby vehicles. To address this issue, Acoustic Vehicle Alerting Systems (AVAS) generate artificial warning sounds when EVs operate at low speeds.

However, conventional AVAS systems typically use fixed sound levels that do not adapt to environmental conditions such as traffic noise or pedestrian density. This motivates the development of an adaptive acoustic alerting system tailored for complex urban environments.

II. Challenges in Indian Urban Roads

Urban traffic environments in India present several challenges:

- High ambient noise levels (70–90 dB)
- Dense pedestrian movement
- Mixed traffic ecosystems (cars, bikes, autos)
- Narrow roads and frequent crossings

These factors require warning systems that dynamically adjust sound intensity and characteristics based on surrounding conditions.

III. System Architecture

The proposed AVAS system integrates environmental sensing, signal processing, and adaptive acoustic generation.

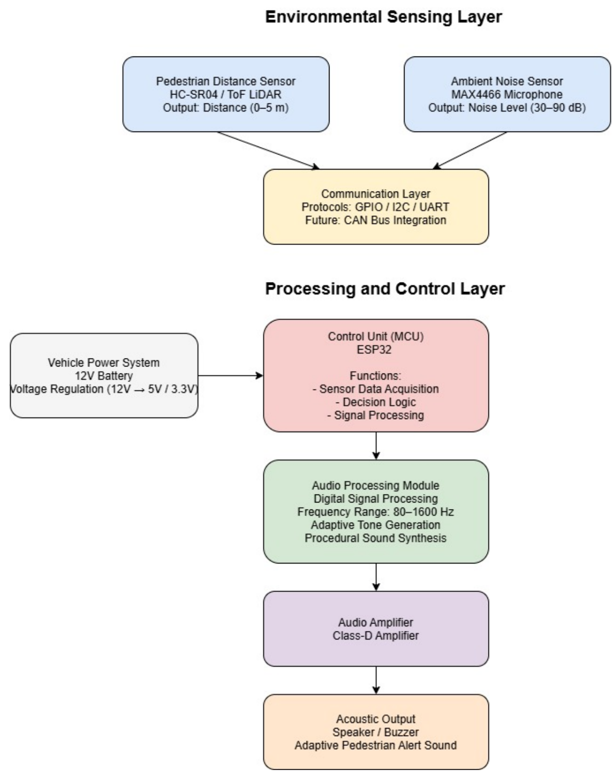

Fig. 1. Proposed AVAS system architecture

Key components include:

- Ambient noise sensor (MAX4466 microphone)
- Pedestrian distance sensor (HC-SR04 ultrasonic sensor)
- Microcontroller (ESP32 / Arduino)
- Audio processing module
- Speaker and amplifier

IV. Decision Algorithm

A rule-based decision algorithm determines alert intensity using pedestrian distance and ambient noise levels.

TABLE I
Rule-Based Alert Decision Model

Distance	Noise Level	Alert Level
<1 m	Any	High
1–3 m	Low	Medium
1–3 m	High	High
>3 m	Low	Low

This approach allows the AVAS sound to adapt dynamically while minimizing unnecessary noise generation.

V. Simulation Environment

To evaluate the system, simulations were performed using MATLAB and Simulink.

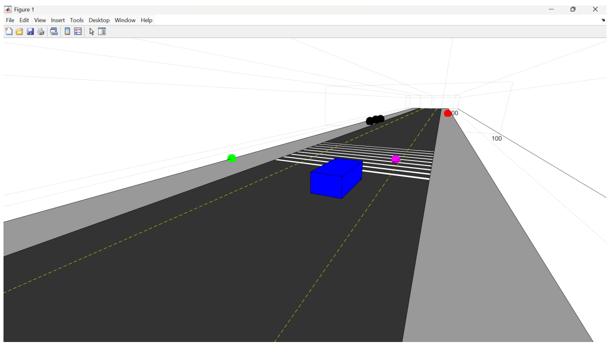

Fig. 2. Driving scenario simulation of the adaptive AVAS system

The simulation models vehicle motion, pedestrian proximity, and environmental conditions.

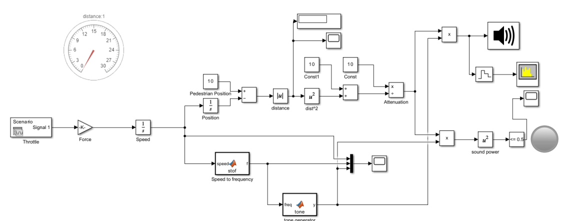

Fig. 3. Simulink model of the adaptive AVAS acoustic signal generation

The Simulink model implements vehicle speed estimation, pedestrian distance calculation, attenuation processing, and tone generation for acoustic alerts. Vehicle speed is mapped to acoustic frequency while the alert intensity is modulated by distance-based attenuation.

The attenuation of sound with pedestrian distance can be approximated as

$$A = \frac{1}{d^2} \tag{1}$$

where d is the distance between the vehicle and the pedestrian.

VI. Hardware Prototype

A hardware prototype was developed to validate the adaptive AVAS concept.

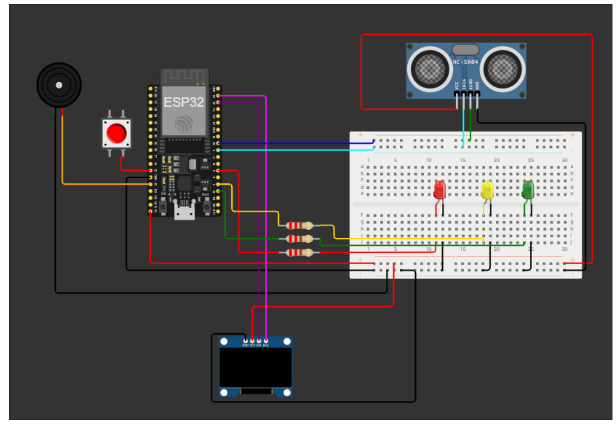

Fig. 4. Hardware prototype implementation

The prototype integrates an ESP32 controller, ultrasonic sensor, and buzzer-based acoustic output.

VII. Results and Observations

Testing of the simulation and prototype showed:

- Alert intensity increases as pedestrian distance decreases
- Higher ambient noise produces stronger alerts
- Adaptive alerts reduce unnecessary sound generation

These observations demonstrate the feasibility of adaptive AVAS systems for urban environments.

VIII. Conclusion

This work presents an adaptive acoustic vehicle alerting system designed for Indian urban traffic conditions. By integrating environmental sensing and rule-based decision logic, the system dynamically adjusts warning sounds according to pedestrian proximity and ambient noise levels. Simulation and prototype validation demonstrate the effectiveness of adaptive AVAS signals for improving pedestrian awareness.

References

[1] European Commission, “Acoustic Vehicle Alerting System (AVAS) Requirements for Electric Vehicles,” Regulation No. 138, 2019.

[2] SAE International, “Minimum Sound Requirements for Hybrid and Electric Vehicles,” SAE Standard J2889, 2011.

[3] Siemens, “Acoustic Vehicle Alerting System for Electric Vehicles,” Technical White Paper.

[4] Technical University of Munich, “Simulation and Design of Acoustic Vehicle Alerting Systems for Electric Vehicles,” Research Report.

[5] ANSYS Blog, “Acoustic Vehicle Alerting Systems for Electric Vehicles,” ANSYS Engineering Simulation Blog.

[6] J. Blauert, *Spatial Hearing: The Psychophysics of Human Sound Localization*. MIT Press, 1997.

[7] GitHub Repository, “AVAS Simulation Project,” Available: https://github.com

[8] GitHub Repository, “Highway Vehicle Sound Dataset,” Available: https://github.com

[9] YouTube Demonstration, “Electric Vehicle Acoustic Alerting System Demonstration.”